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SDN-Driven Dynamic Traffic Slicing and Anti-Starvation Scheduling

Mini-project presentation

Computer Networks Laboratory (CSE20110)

Department of Computer Engineering and Technology

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Group 2 Panel H Batch H2

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Academic Year: 2026–2027 | Presentation date: 22 October 2026

Problem and Objectives

Problem

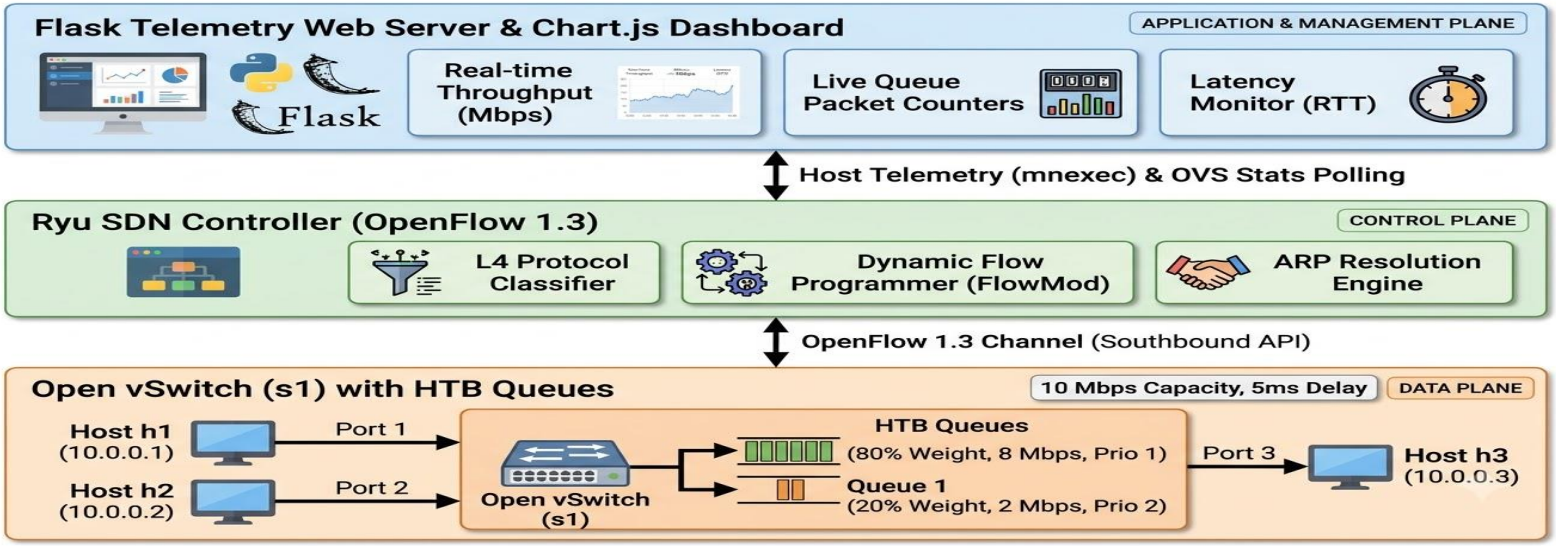
FIFO scheduling can delay interactive packets during congestion, causing high latency and packet loss.

Strict Priority can improve interactive traffic but may starve bulk TCP transfers.

Objectives

1. Classify interactive ICMP/UDP and bulk TCP traffic using OpenFlow 1.3.
2. Configure two queues with an 80:20 allocation on a 10 Mbps bottleneck.
3. Reduce latency and packet loss while preserving service for bulk traffic.

Proposed Design and Tools



Tools: Mininet, Open vSwitch (OVS), Ryu, OpenFlow 1.3, Linux Traffic Control (tc), Flask andChart.js.

Protocols: OpenFlow 1.3 for controller–switch rules; ARP for address resolution; ICMP/UDP and TCP for test traffic.

Addressing: 10.0.0.0/24; h1 10.0.0.1, h2 10.0.0.2, h3 10.0.0.3; controller TCP port 6633.

Implementation and Working

```
mininet@mininet-vm:~$ cd ~/sdn-wfq-project
mininet@mininet-vm:~/sdn-wfq-project$ ryu-manager qos_controller.py
loading app qos_controller.py
loading app ryu.controller.ofp_handler
instantiating app qos_controller.py of QoSController
instantiating app ryu.controller.ofp_handler of OFPHandler
Switch connected. Table-miss entry installed.
[Classifier] Routed 10.0.0.1 (proto 1) -> Queue 0
[Classifier] Routed 10.0.0.2 (TCP Bulk) -> Queue 1
[Classifier] Routed 10.0.0.1 (proto 17) -> Queue 0
[Classifier] Routed 10.0.0.1 (proto 17) -> Queue 0
[Classifier] Routed 10.0.0.2 (TCP Bulk) -> Queue 1
[Classifier] Routed 10.0.0.1 (proto 17) -> Queue 0
[Classifier] Routed 10.0.0.2 (TCP Bulk) -> Queue 1
^[[3~^Z
[1]+  Stopped                 ryu-manager qos_controller.py
mininet@mininet-vm:~/sdn-wfq-project$ ryu-manager qos_controller.py
loading app qos_controller.py
loading app ryu.controller.ofp_handler
instantiating app qos_controller.py of QoSController
instantiating app ryu.controller.ofp_handler of OFPHandler
Switch connected. Table-miss entry installed.
[Classifier] Routed 10.0.0.1 (proto 1) -> Queue 0
[Classifier] Routed 10.0.0.2 (TCP Bulk) -> Queue 1
[Classifier] Routed 10.0.0.1 (proto 17) -> Queue 0
```

Implementation: Create a three-host Mininet topology; configure the 10 Mbps bottleneck and HTB queues; run the Ryu controller and install flow rules.

Working: ICMP/UDP is assigned to Queue 0 (8 Mbps); TCP is assigned to Queue 1 (2 Mbps minimum allocation). Queue statistics and RTT are monitored.

Troubleshooting: Verify controller connectivity, OpenFlow 1.3, queue setup and interface names before testing.

Testing and Results



Test: Compare unmanaged FIFO with SDN traffic slicing, then stress-test bulk TCP traffic.

Expected: Lower interactive latency and packet loss, with bulk traffic continuing to receive bandwidth.

Observed: Average RTT 23.31 → 6.53 ms; maximum RTT 68.61 → 9.60 ms; interactive packet loss 18% → 0%. Bulk TCP reached 5.75 Mbps in the stress test.

Working Demonstration

Function to demonstrate

Live traffic classification, queue allocation and performance monitoring

1. Launch Mininet and start the Ryu controller.
2. Generate ICMP/UDP traffic and a bulk TCP iperf transfer.
3. Compare RTT, packet loss, queue counters and throughput on the dashboard.

Success criterion: Interactive traffic remains responsive while bulk TCP continues receiving bandwidth.

Conclusion and Contributions

Outcome: SDN traffic slicing reduced average RTT and interactive packet loss in the tested virtual network.

Application: QoS support for interactive applications alongside file transfers.

Limitation: Results are from a controlled Mininet setup; larger networks and more workloads need testing.

Future scope: Multi-subnet SDN routing, repeated trials and Jain's Fairness Index.

Team contributions:
Nilam Vanjani & Bhavika Lalwani – project design, implementation, testing and report preparation.

AI use: AI tools were used for language editing and report organization.

References

[1] N. McKeown et al., “OpenFlow: Enabling Innovation in Campus Networks,” ACM SIGCOMM CCR, vol. 38, no. 2, 2008.

[2] Open Networking Foundation, OpenFlow Switch Specification, Version 1.3.0, 2012.

[3] Mininet Documentation — https://mininet.org/

[4] Open vSwitch Documentation — https://docs.openvswitch.org/

[5] Ryu SDN Framework — https://ryu.readthedocs.io/

[6] Linux Traffic Control Documentation — https://docs.kernel.org/networking/

Figures and reported results are taken from the project report. Performance values apply to the tested virtual network and workloads.